

Main goal of this project is to fine-tune a pre-trained BERT model for sequence classification, specifically for sentiment analysis, using the IMDB dataset

```
In [43]: %pip install tensorflow==2.16.1
          %pip install transformers
```

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Requirement already satisfied: tensorflow==2.16.1 in
/usr/local/lib/python3.12/dist-packages (2.16.1)
Requirement already satisfied: absl-py>=1.0.0 in
/usr/local/lib/python3.12/dist-packages (from tensorflow==2.16.1)
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Requirement already satisfied: astunparse>=1.6.0 in
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Requirement already satisfied: gast!=0.5.0,!0.5.1,!0.5.2,>=0.2.1
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tensorflow==2.16.1) (0.7.0)
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Requirement already satisfied:
protobuf!=4.21.0,!4.21.1,!4.21.2,!4.21.3,!4.21.4,!4.21.5,
<5.0.0dev,>=3.20.3 in /usr/local/lib/python3.12/dist-packages
(from tensorflow==2.16.1) (4.25.9)
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/usr/local/lib/python3.12/dist-packages (from tensorflow==2.16.1)
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Requirement already satisfied: setuptools in
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Requirement already satisfied: wheel<1.0,>=0.23.0 in
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Requirement already satisfied: charset_normalizer<4,>=2 in
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/usr/local/lib/python3.12/dist-packages (from requests<3,>=2.21.0-
>tensorflow==2.16.1) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
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>tensorflow==2.16.1) (2026.5.20)
Requirement already satisfied: markdown>=2.6.8 in
/usr/local/lib/python3.12/dist-packages (from
tensorboard<2.17,>=2.16->tensorflow==2.16.1) (3.10.2)
Requirement already satisfied: tensorboard-data-

server<0.8.0,>=0.7.0 in /usr/local/lib/python3.12/dist-packages
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Requirement already satisfied: werkzeug>=1.0.1 in
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Requirement already satisfied: transformers in
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Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in
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Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in
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hub<2.0,>=1.3.0->transformers) (1.5.0)
Requirement already satisfied: httpx<1,>=0.23.0 in
/usr/local/lib/python3.12/dist-packages (from huggingface-
hub<2.0,>=1.3.0->transformers) (0.28.1)

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hub<2.0,>=1.3.0->transformers) (4.15.0)
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>huggingface-hub<2.0,>=1.3.0->transformers) (4.13.0)
Requirement already satisfied: certifi in
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>huggingface-hub<2.0,>=1.3.0->transformers) (2026.5.20)
Requirement already satisfied: httpcore==1.* in
/usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0-
>huggingface-hub<2.0,>=1.3.0->transformers) (1.0.9)
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>huggingface-hub<2.0,>=1.3.0->transformers) (3.15)
Requirement already satisfied: h11>=0.16 in
/usr/local/lib/python3.12/dist-packages (from httpcore==1.*-
>httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers)
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Requirement already satisfied: click>=8.2.1 in
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>huggingface-hub<2.0,>=1.3.0->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in
/usr/local/lib/python3.12/dist-packages (from typer>=0.20.0-
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Requirement already satisfied: markdown-it-py>=2.2.0 in
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>typer>=0.20.0->huggingface-hub<2.0,>=1.3.0->transformers) (4.2.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in
/usr/local/lib/python3.12/dist-packages (from rich>=13.8.0-
>typer>=0.20.0->huggingface-hub<2.0,>=1.3.0->transformers)
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Requirement already satisfied: mdurl~=0.1 in
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py>=2.2.0->rich>=13.8.0->typer>=0.20.0->huggingface-
hub<2.0,>=1.3.0->transformers) (0.1.2)

```

```
In [44]: from transformers import BertTokenizer
```

BERT model itself doesn't "assign" the IDs on the fly; it uses a fixed vocabulary that was learned during its extensive pre-training on a massive amount of text data. The tokenizer's job is to efficiently map new text to this existing vocabulary and its associated IDs

BERT uses a technique called WordPiece tokenization. Here's how it works when it encounters a "new" word:

Subword Segmentation: Instead of only storing whole words, the tokenizer's vocabulary also contains common subword units (prefixes, suffixes, roots, etc.). When it sees a word not in its full-word vocabulary, it tries to break that word down into smaller subword units that are in its vocabulary. For example, if 'unbelievable' isn't in the vocabulary, it might break it into 'un', '##believe', and '##able'. The '##' indicates that it's a continuation of a previous token.

[UNK] Token for Truly Unknown Parts: If a part of a word (or a whole word) cannot be broken down into any known subword units from its vocabulary, the tokenizer will replace that part with a special token called [UNK] (for 'unknown'). This ensures that the model always receives a numerical ID for every part of the input, even if it's an unfamiliar word.

This approach has two main benefits:

Handles Out-of-Vocabulary (OOV) words: The model doesn't just crash or ignore new words; it can still process them by understanding their constituent parts.

Manages Vocabulary Size: Instead of needing an impossibly large vocabulary for every possible word, a more compact vocabulary of common words and subword units can cover a vast range of language.

```
In [45]: tokenizer = BertTokenizer.from_pretrained('bert-base-uncased')
```

```
In [45]:
```

```
In [46]: words = ['parachute', 'paraglide', 'scubadiving', 'scubadiver', 'scubadive']
```

```
In [47]: for i in words:
          tok_word = tokenizer.tokenize(i)
          print(f'{i} tokenized to {tok_word}')
```

```
parachute tokenized to ['parachute']
paraglide tokenized to ['para', '##gli', '##de']
scubadiving tokenized to ['scuba', '##di', '##ving']
scubadiver tokenized to ['scuba', '##di', '##ver']
scubadive tokenized to ['scuba', '##di', '##ve']
```

```
In [48]: from datasets import load_dataset

ds = load_dataset("stanfordnlp/imdb")
ds
```

```
Out[48]: DatasetDict({
  train: Dataset({
    features: ['text', 'label'],
    num_rows: 25000
  })
  test: Dataset({
    features: ['text', 'label'],
    num_rows: 25000
  })
  unsupervised: Dataset({
    features: ['text', 'label'],
    num_rows: 50000
  })
})
```

```
In [48]:
```

```
In [49]: def tokenize_function(examples):
          return tokenizer(examples['text'], truncation = True, padding =
            'max_length')
```

```
In [50]: tokenized_dataset = ds.map(tokenize_function, batched = True)
```

```
Map:   0%|          | 0/25000 [00:00<?, ? examples/s]
```

We need to import the BertForSequenceClassification class from the transformers library before we can use it

```
In [51]: from transformers import BertForSequenceClassification

model = BertForSequenceClassification.from_pretrained('bert-base-uncased', num_labels = 2)
```

```
Loading weights: 0%|          | 0/199 [00:00<?, ?it/s]
```

```
BertForSequenceClassification LOAD REPORT from: bert-base-uncased
```

Key	Status
cls.predictions.transform.dense.weight	UNEXPECTED
cls.predictions.bias	UNEXPECTED
cls.predictions.transform.LayerNorm.weight	UNEXPECTED
cls.predictions.transform.dense.bias	UNEXPECTED
cls.seq_relationship.bias	UNEXPECTED
cls.predictions.transform.LayerNorm.bias	UNEXPECTED
cls.seq_relationship.weight	UNEXPECTED
classifier.weight	MISSING
classifier.bias	MISSING

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.

```
In [52]: from transformers import TrainingArguments

training_args = TrainingArguments(
    output_dir="sentiment_model",
    # evaluation_strategy="epoch", # Temporarily removed to fix
    # TypeError, consider updating transformers to re-enable.
    save_strategy="epoch",
    learning_rate=2e-5,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    num_train_epochs=1,
    weight_decay=0.01,
    logging_steps=10,
)
```

```
In [53]: from transformers import Trainer
import numpy as np
from sklearn.metrics import accuracy_score, f1_score,
precision_recall_fscore_support

def compute_metrics(eval_pred):
    logits, labels = eval_pred
    predictions = np.argmax(logits, axis=-1)
    # For binary classification
    return {"accuracy": accuracy_score(labels, predictions)}

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=tokenized_dataset['train'].shuffle(seed
=42).select(range(2000)),
    eval_dataset=tokenized_dataset['test'].shuffle(seed
=42).select(range(500)), # Added a comma here
    compute_metrics=compute_metrics,
)
```



```
In [54]: trainer.train()
```



Step	Training Loss
10	0.696395
20	0.695855
30	0.634305
40	0.556431
50	0.460099
60	0.381289
70	0.296539
80	0.230475
90	0.297172
100	0.258325
110	0.322434
120	0.331690
130	0.352652
140	0.266500
150	0.323551
160	0.341851
170	0.361478
180	0.263020
190	0.323439
200	0.417075
210	0.306405
220	0.333825
230	0.122668
240	0.243280
250	0.417892

Writing model shards: 0% | 0/1 [00:00<?, ?it/s]

TrainOutput(global_step=250, training_loss=0.3693858208656311, metrics={'train_runtime': 230.7195, 'train_samples_per_second':

```
8.669, 'train_steps_per_second': 1.084, 'total_flos':  
526222110720000.0, 'train_loss': 0.3693858208656311, 'epoch':  
1.0})
```

In [55]: `trainer.evaluate()`

 [63/63 00:15]

```
{'eval_loss': 0.2856241464614868,  
 'eval_accuracy': 0.896,  
 'eval_runtime': 15.558,  
 'eval_samples_per_second': 32.138,  
 'eval_steps_per_second': 4.049,  
 'epoch': 1.0}
```

In [55]:

In [56]:

```
import pandas as pd  
  
small_eval_dataset =  
tokenized_dataset['test'].shuffle(seed=42).select(range(500))  
  
predictions_output = trainer.predict(small_eval_dataset)  
  
raw_logits = predictions_output.predictions  
true_labels = predictions_output.label_ids  
predicted_labels = np.argmax(raw_logits, axis=1)  
  
from scipy.special import softmax  
probabilities = softmax(raw_logits, axis=1)  
confidence_scores = np.max(probabilities, axis=1)  
  
results_df = pd.DataFrame({  
    "Review_Text": small_eval_dataset["text"],  
    "True_Label": true_labels,  
    "Predicted_Label": predicted_labels,  
    "Confidence": confidence_scores  
})
```

In [57]: results_df

Out[57]:

	Review_Text	True_Label	Predicted_Label	Confidence
0	 When I unsuspectedly rented A Thou...	1	1	0.981062
1	This is the latest entry in the long series of...	1	1	0.691370
2	This movie was so frustrating. Everything seem...	0	0	0.986099
3	I was truly and wonderfully surprised at "O' B...	1	1	0.952280
4	This movie spends most of its time preaching t...	0	0	0.979565
...	...	...	...	...
495	I'm not going to say too much as this movie is...	0	0	0.985824
496	Definitely the worst movie I have ever seen in...	0	0	0.988208
497	Wow, there are no words to describe how bad th...	0	0	0.986613
498	Gritty drama? Emotionally powerful? Blah! The ...	0	0	0.987564
499	The CinemaScope color cinematography of Leon S...	0	0	0.538125

500 rows × 4 columns

In [57]:

